
Part 107 Cheat Sheet

Quick Reference Guide for Exam Day

Every critical fact, number, and rule you need to know for the FAA Part 107 Knowledge Test, organized by topic for fast review. Print this out and review it right before your exam.

375

Key Facts

20

Topics

100%

Exam Coverage

Quick-Ref

Format

FLYCENSED

flycensed.com

REGULATIONS

Part 107 Core Regulations

> What is Part 107?

- Part 107 covers drones 0.55-55 pounds
- Established in 2016
- Applies to commercial operations
- Requires remote pilot certification

> Basic Operating Limitations

- 400 feet AGL maximum altitude
- 100 mph maximum airspeed
- 3 statute mile visibility minimum
- 55 pounds maximum weight

> Remote Pilot Certification

- 60-question knowledge test
- 70% passing score required
- Certificate valid 3 years
- Online recurrent training every 24 months (FREE, no exam)
- 16 years minimum age

> Key Operational Rules

- Visual observer required, separate from pilot
- Only one remote pilot can be in command
- Daylight operations only without waiver
- Preflight inspection required
- 8-hour minimum after alcohol

> Airspace and Authorization

- LAANC required in controlled airspace
- Check TFRs before every flight
- Cannot fly in Prohibited/Restricted areas
- Class G airspace may be unrestricted
- Airspace authorization takes 5-10 minutes

WEATHER

Weather Fundamentals

> Weather Elements and Flight Safety

- Wind affects control and position
- Rain reduces visibility and damages drones
- Visibility minimum: 3 statute miles
- Stay 500 feet below clouds

- Cold reduces battery performance

> Wind and Its Effects

- Headwind increases control authority
- Tailwind reduces flight time
- Crosswind causes drift
- Wind from surface to 400 feet varies
- Check METAR for wind data

> Visibility and Atmospheric Conditions

- Visibility minimum: 3 statute miles
- Fog reduces to <1/2 SM
- Mist/haze: 1/2-6 SM
- Check METAR for current visibility
- TAF shows forecast changes

> Clouds and Ceiling Altitude

- Clouds reported in eighths
- FEW=1-2/8, SCT=3-4/8, BKN=5-7/8, OVC=8/8
- Stay 500 feet below
- Ceiling = lowest BKN/OVC layer
- Low clouds = poor visibility

PERFORMANCE

Density Altitude & Performance

> What is Density Altitude?

- Combines altitude, temperature, humidity
- Hot, humid day = high DA
- Cold day = low DA
- Affects aircraft performance
- Calculate from temperature and pressure

> Factors Affecting Density Altitude

- Higher altitude = higher DA
- Higher temperature = higher DA
- Higher humidity = slightly higher DA
- Higher pressure = lower DA
- Multiple factors combine

> Performance Effects and Calculations

- High DA reduces performance
- DA calculator online available
- Affects climb rate most
- Takeoff distance increases
- Hot + high alt + humid = worst

OPERATIONS

Aeronautical Decision Making

> What is Aeronautical Decision Making?

- Systematic decision process
- Most accidents from poor decisions
- Applies to all flight decisions
- Requires analyzing options
- Part of pilot training

> The DECIDE Model

- Six-step structured process
- Detect problem first
- Estimate severity
- Choose action deliberately
- Evaluate result after

> Risk Assessment and Mitigation

- Identify all risks
- Develop mitigation for each
- Residual risk must be acceptable
- When unacceptable, cancel
- Conservative approach best

REGULATIONS

Registration & Remote ID

> Drone Registration Requirements

- Registration fee: \$5
- Valid 3 years
- Applies to drones 0.55+ pounds
- Must mark registration number on aircraft
- Owner must be 13+ years old

> Remote ID Requirements

- All drones must broadcast Remote ID
- Shows location and operator ID
- Network or broadcast capability
- Exemptions for aircraft under 0.55 lbs
- Enables FAA compliance tracking

> Operator Registration and Tracking

- Maintain flight logs with date, time, location
- Document all incidents

- Keep 24-month record retention
- Certificate is primary operator authorization
- Documentation aids insurance and compliance

> Liability and Insurance Considerations

- Insurance not required but highly recommended
- Standard coverage: \$1-2 million
- Covers property damage and bodily injury
- Costs vary by coverage and risk
- Professional operators carry insurance

WEATHER

Reading METARs

> METAR Format and Elements

- Hourly reports from airports
- Format: KJFK 121251Z...
- Wind first (direction/speed)
- Visibility in SM
- Multiple layers of clouds

> Wind and Visibility Codes

- Wind: direction + speed (knots)
- Gusts appended with G notation
- Variable wind: VRB
- Visibility in statute miles
- Part 107 needs 3SM minimum

> Weather Phenomena Codes

- Intensity: - (light), none (moderate), + (heavy)
- RA=rain, SN=snow, DZ=drizzle
- FG=fog, BR=mist, HZ=haze
- TS=thunderstorm
- FZ prefix=freezing conditions

> Clouds, Temperature, and Altimeter

- Cloud layers reported separately
- T25/15 = 25°C temp, 15°C dew point
- Small dew point spread = high moisture
- Altimeter: A2990
- A>2992 = high pressure
- A<2992 = low pressure

PERFORMANCE

Weight, Balance & Loading

> Weight and Balance Principles

- CG must be in envelope
- CG too far forward = nose-heavy
- CG too far aft = unstable
- MTOW = 55 pounds
- Part 107 weight limit cannot be waived

> Calculating Center of Gravity

- Moment = weight × distance
- CG = total moment ÷ total weight
- Aircraft CG envelope given
- Calculate loaded CG before flight
- Must be in allowable range

> Loading and Payload Considerations

- Payload location affects CG
- Front load = CG forward
- Rear load = CG aft
- Calculate CG with all payload
- Test unfamiliar configurations

OPERATIONS

Crew Resource Management

> What is Crew Resource Management?

- Uses all available resources
- Prevents miscommunication
- Pilot welcomes crew input
- Crew speaks up about concerns
- Manages human limitations

> Communication and Coordination

- Two-way communication essential
- Briefing before flight
- Continuous information exchange
- Standardized terminology
- Acknowledge information received

> Error Management and Challenge

- Errors will happen
- Catch before accidents occur
- Cross-check information
- Challenge/response procedures
- No-blame culture

REGULATIONS

Waivers & Special Operations

> What is a Part 107 Waiver?

- Authorization to deviate from Part 107 rules
- FAA uses risk-based analysis
- Issued for specific operations
- 55 lbs, 400 ft, 100 mph cannot be waived
- Typically valid 2 years

> Requesting a Waiver

- Submit through FAA online system
- 90-day advance notice recommended
- Must justify safety measures
- FAA may add conditions to approval
- Approval is not guaranteed

> Common Waiver Categories

- Night ops NO LONGER need a waiver (since 2021)
- BVLOS is most common waiver now
- Over-people has 4 categories (some no waiver needed)
- Cannot waive the 55-pound weight limit
- Must demonstrate equivalent safety level

> Waiver Conditions and Compliance

- Waiver conditions are mandatory
- Brief all crew on conditions
- Violating conditions = serious violation
- Document compliance
- Carry printed waiver at flight site

WEATHER

Reading TAFs

> TAF Format and Purpose

- Issued 4-5 times daily
- Valid 24-30 hours
- Format similar to METAR
- FM = new period
- BECMG = gradual change
- TEMPO = temporary

> TAF Interpretation and Change Groups

- Base forecast initial conditions
- FM = new forecast period
- BECMG = gradual change 1-2 hrs
- TEMPO = temporary conditions
- Time format: DDHR (day/time UTC)

> Using TAFs for Flight Planning

- Check for FM changes during operation time
- Note BECMG time windows
- TEMPO may cause operation delays
- TAFs are forecasts only
- Verify with current METAR before launch

> TAF Code Examples

- Wind: direction and speed
- Visibility always included
- Cloud layer with altitude
- FM indicates time change
- TEMPO indicates brief poor conditions

OPERATIONS

Physiology: Alcohol, Drugs & Fatigue

> Alcohol and Flight Safety

- Impairs judgment and coordination
- BAC limit: 0.04%
- 8-hour rule from drinking
- One drink per hour metabolizes
- Never fly while intoxicated

> Drugs and Medications

- Illegal drugs prohibited
- Marijuana prohibited
- Prescription meds require check
- OTC meds can be problematic
- Impairment = not flying

> Fatigue and Alertness

- Fatigue = impaired judgment
- One bad night affects performance
- Circadian rhythm important
- 8 hours sleep minimum
- Tired pilot = don't fly

REGULATIONS

Night Operations & Ops Over People

> Night Operations Rules

- Night ops allowed WITHOUT waiver since April 2021
- Anti-collision lighting must be visible for 3+ statute miles
- Pilot must have completed training after April 6, 2021
- Daylight = 30 min before sunrise to 30 min after sunset
- Civil twilight is included in daylight hours

> Night Flight Considerations

- Spatial disorientation risk increases
- Visual observer less effective
- Ground lighting can cause disorientation
- Obstacles invisible at night
- Additional spotters recommended

> Operations Over People - Categories and Requirements

- Category 1: under 0.55 lbs, no auth needed
- Category 2: no sharp/rotating parts, design tested
- Category 3: Cat 2 for assemblies, needs Remote ID
- Category 4: airworthiness certified, Remote ID + 50mph limit
- Aircraft characteristics determine category

> Over-People Operations Safety and Mitigation

- Detailed risk assessment required
- Insurance essential
- Multiple spotters needed
- Communication protocols established
- Contingency procedures for emergencies

WEATHER

Thunderstorms & Hazardous Weather

> Thunderstorm Formation and Stages

- Developing, Mature, Dissipating stages
- Mature stage most dangerous
- Requires instability, moisture, lifting
- Stay 10nm from thunderstorms
- Supercells produce tornadoes

> Hazards Within Thunderstorms

- Severe turbulence causes damage
- Hail damages aircraft
- Lightning is lethal
- Wind shear causes loss of control

- Microbursts = extreme downwind

> Identification and Avoidance

- TS in METAR indicates activity
- Dark towering clouds indicate storms
- Lightning = positive identification
- 10nm horizontal separation
- Avoid storm areas entirely

> Other Hazardous Weather Phenomena

- Wind shear on approaches
- Icing possible in cold clouds
- Mountain wave on lee slopes
- Fog reduces visibility to zero
- Dust storms = zero visibility

OPERATIONS

Emergency Procedures

> What Constitutes an Emergency?

- Loss of control = emergency
- Comm loss = emergency
- Battery critical = emergency
- Weather deterioration = emergency
- Immediate action required

> Emergency Procedures and Recovery

- Stabilize aircraft first
- Identify problem second
- Apply recovery procedure
- Land if unable to recover
- Contingency plan before flight

> Emergency Equipment and Resources

- First aid kit available
- Fire extinguisher for batteries
- Spare batteries for critical ops
- Emergency contacts listed
- Crew briefing required

AIRSPACE

Airspace Classification Overview

> Seven Classes of Airspace

- Classes A-E are controlled airspace
- Class G is uncontrolled airspace
- Part 107 limited to 400 feet AGL
- Classification varies by location
- Different requirements per class

> **Controlled vs. Uncontrolled Airspace**

- Controlled airspace: A, B, C, D, E
- Uncontrolled: Class G
- ATC services in controlled airspace
- LAANC required in controlled airspace
- Part 107 rules apply everywhere

> **Airspace Dimensions and Altitudes**

- Horizontal and vertical boundaries
- MSL vs. AGL altitudes
- Part 107 limits in AGL
- Class B extends 20-30 nm
- Sectional charts show boundaries

> **Authorization and Communication Requirements**

- Class B requires clearance
- Class C requires communication
- Class D requires tower communication
- LAANC authorizes for drones
- Approval not guaranteed

Class B, C, D Airspace

> **Class B Airspace - Busy Airports**

- Surrounds busiest airports
- Surface to 10,000 feet MSL
- Solid blue lines on chart
- ATC clearance required
- Strict adherence to clearance

> **Class C Airspace - Medium Traffic Airports**

- Medium-traffic airports
- Surface to ~4,000 feet AGL
- Magenta lines on chart
- Two-way radio communication required
- LAANC alternative for drones

> **Class D Airspace - Smaller Airports with Towers**

- Smaller airports with towers
- Surface to 2,500 feet AGL
- Dashed blue lines on chart
- Radio communication required

- Clearance not always required
 - > **Special Considerations for Class B, C, D**
 - Check LAANC 24 hours before
 - Request auth in advance
 - TRSA may provide radar service
 - Weather can change rapidly
 - Denial requires alternative planning
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Class E & G Airspace

- > **Class E Airspace - General Controlled Airspace**
 - Controlled airspace, non-major airports
 - Dashed magenta on chart
 - Surface or 700/1200 feet floor
 - Upper Class E above operations
 - LAANC may be required
 - > **Class G Airspace - Uncontrolled**
 - Uncontrolled airspace
 - Depicted in white on chart
 - No ATC services
 - Default airspace classification
 - Part 107 rules still apply
 - > **Differences Between Controlled and Uncontrolled**
 - Controlled = ATC separation
 - Uncontrolled = see-and-avoid
 - Controlled = stricter minimums
 - Class E = ATC service available
 - Class G = pilot responsibility
 - > **Finding Your Airspace Class**
 - Sectional charts show all airspace
 - Online tools available
 - Color coding indicates class
 - Contact Flight Service if unsure
 - Check for TFRs and NOTAMs
-

Special Use Airspace & TFRs

- > **Special Use Airspace Overview**
 - SUA marked on sectional charts
 - Restricted = non-military hazards
 - Prohibited = no flight allowed
 - Warning = off coasts
 - Avoid unless authorized
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> **Military Training Routes (MTRs)**

- Marked on sectional charts
- Low-level training routes
- VR = visual, IR = instrument
- Check NOTAMs for activity
- Avoid during active training

> **Military Operations Areas (MOAs)**

- Flight not prohibited
- Operations at pilot's risk
- Marked on sectional charts
- Check NOTAMs for activity
- ATC coordination available

> **Temporary Flight Restrictions (TFRs)**

- Checked before every flight
 - Check faa.gov/tfr website
 - Violations = serious penalties
 - Can change with short notice
 - Typical radius: 10nm from point
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Reading Sectional Charts

> **Chart Overview and Purpose**

- Detailed aviation maps
- Updated every 6 months
- Covers ~85x60 nautical miles
- 37 charts for continental US
- Free online from FAA

> **Airspace Depiction on Charts**

- Blue = Class B,D
- Magenta = Class C,E
- White = Class G
- Numbers = altitude in hundreds of feet
- Special areas clearly marked

> **Airports and Runways**

- Circle = uncontrolled airport
- Tower symbol = controlled
- Runway heading labeled
- Elevation in feet MSL
- Tower frequency in blue

> **Navigation Aids and Obstacles**

- VORs with frequency and variation
- Obstacles with heights labeled
- Elevation contours show terrain

- Cities and towns marked
 - Power lines indicated
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LAANC & Airspace Authorization

> What is LAANC?

- FAA authorization system for drones
- Available in most controlled airspace
- Real-time approval system
- Mobile app access
- 24/7 availability

> How LAANC Works

- Enter operation details in app
- System checks compatibility
- Approval in minutes
- Denials show conflict reason
- Authorization has time window

> LAANC Coverage and Limitations

- Available nationwide but not complete
- Major airports well covered
- Coverage varies by location
- Alternatives exist for gaps
- Phone contact to ATC possible

> Pre-Flight Airspace Authorization Checklist

- Check airspace class first
 - Request LAANC 24+ hours prior
 - Check TFR database
 - Review NOTAMs
 - Keep authorization on site
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Quick Numbers Reference

Max Altitude	400 ft AGL
Max Speed	100 mph / 87 knots
Max Weight	55 lbs (25 kg)
Min Visibility	3 statute miles
Cloud Clearance	500 ft below, 2000 ft horizontal
Min Pilot Age	16 years old
Certificate Valid	3 years
Recurrent Training	Every 24 calendar months (FREE)
Registration Fee	\$5 for 3 years
Knowledge Test	60 questions, 70% to pass, 2 hours
Alcohol Rule	8 hours, 0.04% BAC max
Anti-Collision Light	Visible 3+ statute miles (night)
Waiver Lead Time	90 days recommended
LAANC	Required in Class B/C/D/E airspace
Incident Report	Within 10 days to FAA